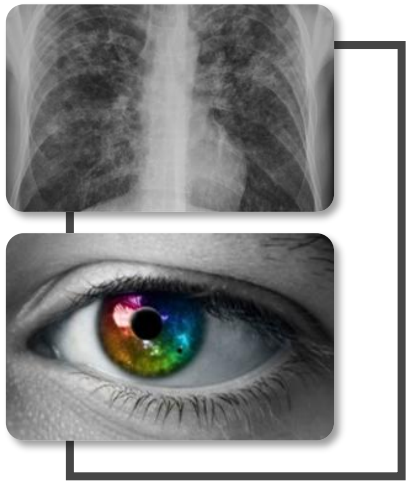
Medi Scan

YOUR HEALTH ALLY

A DEEP LEARNING TOOL FOR TUBERCULOSIS AND EYE DISEASE PREDICTION USING CHEST X-RAYS AND RETINAL IMAGES



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OVERVIEW

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- ✓ Aim and Objectives
- ✓ Materials and Methodology
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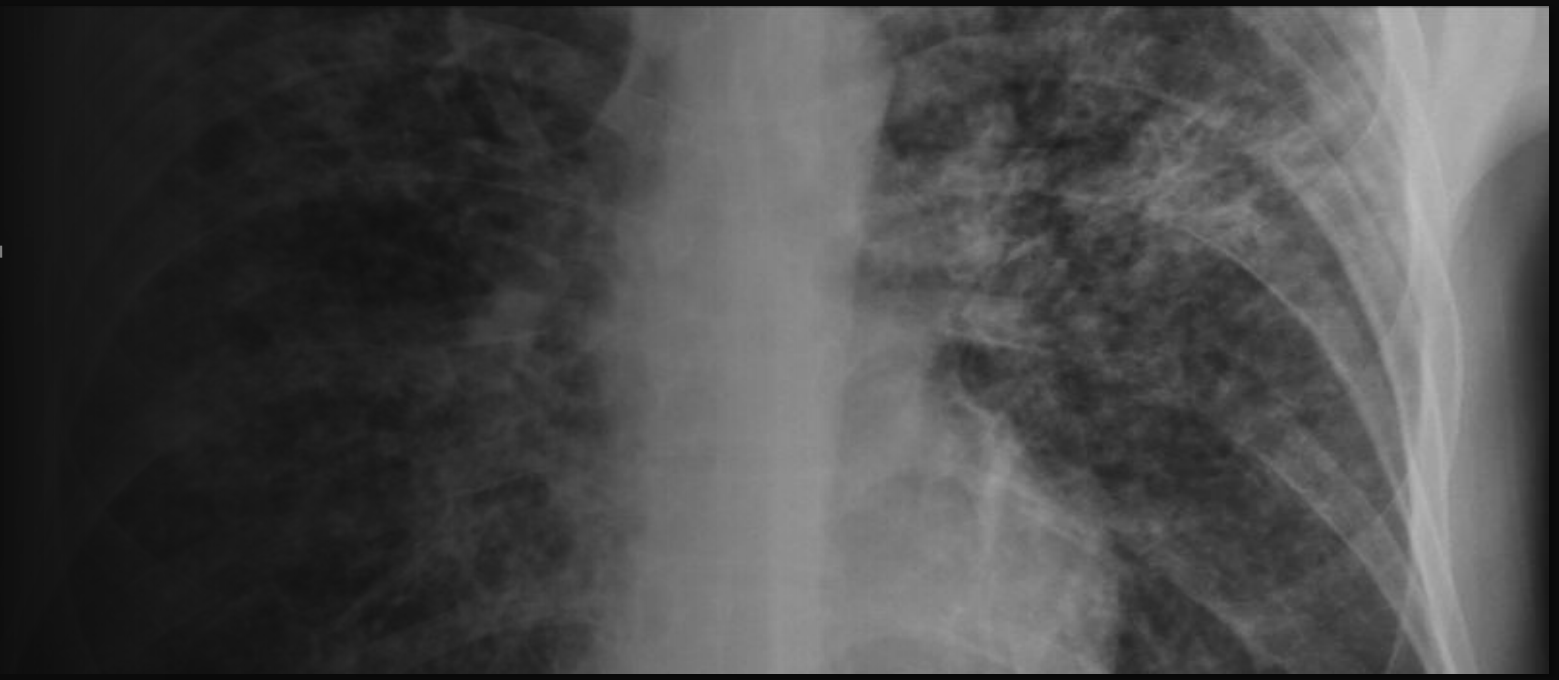


ABSTRACT

- **Web-based tool:** MediScan utilizes deep learning for disease detection.
- **Analyzes medical images:** Chest X-rays for T.B. and retinal images for eye diseases.
- **Promising accuracy:** 93.8% for T.B. and 87% for eye disease.
- **Empowering individuals:** Enables self-assessment and proactive healthcare management.
- **Bridges the AI gap:** Creates a user-friendly tool from complex technology.
- **Improved health outcomes:** Particularly in resource-limited settings.

INTRODUCTION

Rising Healthcare Burden

- The global healthcare burden is rising, demanding innovative solutions for early disease detection and intervention.
- Early disease detection is crucial for effective treatment, better health outcomes and reduced costs.
- Traditional methods can be:
  - Expensive & Resource-intensive
  - Time-consuming & Invasive
  - Limited Accessibility & Accuracy

Introducing MediScan

- MediScan, a medical imaging software tool that utilizes deep learning CNNs for disease prediction.
- Analyzes chest X-rays and retinal images for tuberculosis and eye diseases like cataracts, glaucoma, diabetic retinopathy.
- MediScan offers an accessible and a non-invasive approach to disease detection.

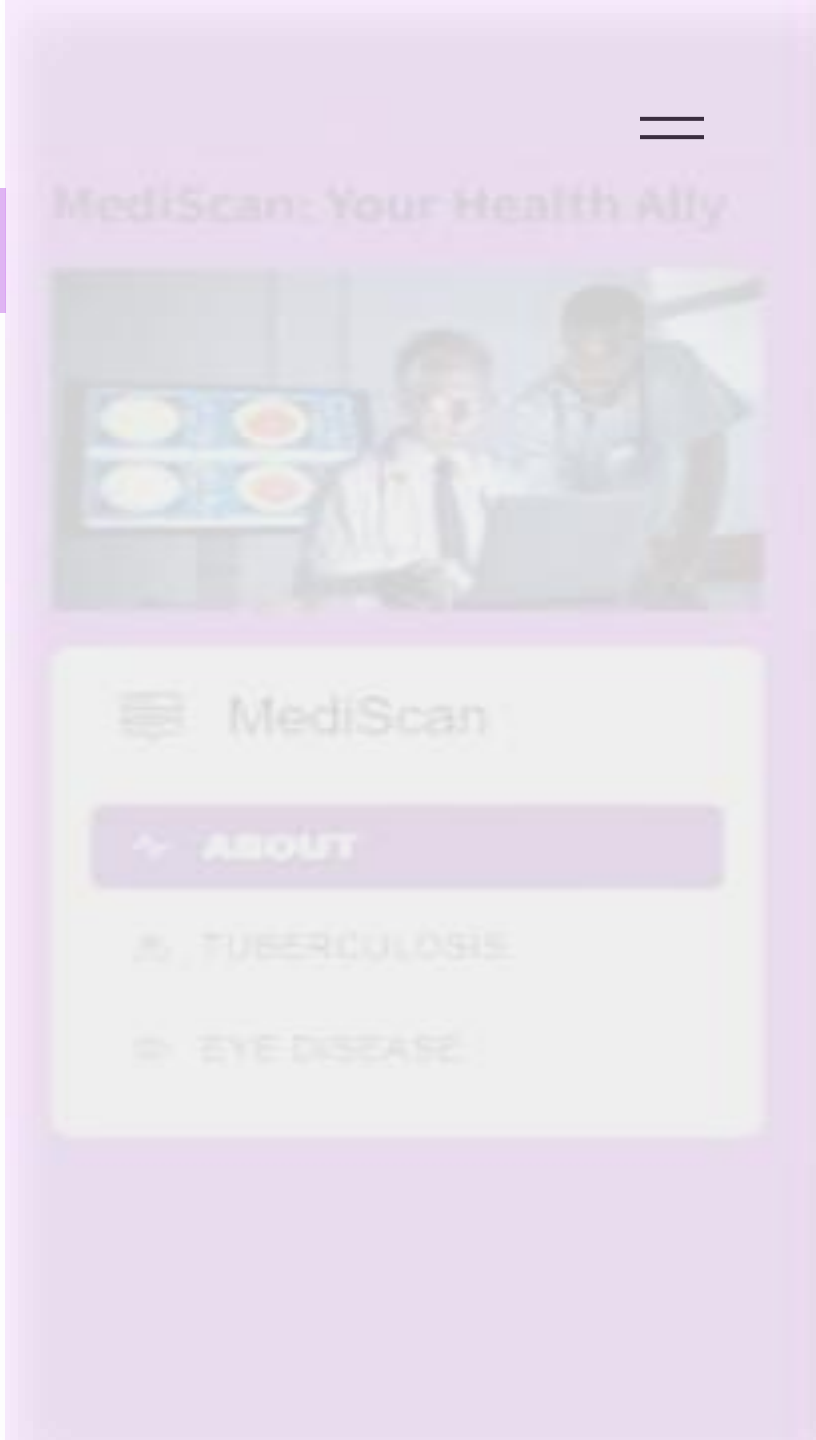
FUNCTIONALITIES OF MEDISCAN

Deep learning architecture: Powered by Keras Sequential API and VGG19 model.

Analyzes medical images: Detects T.B. from chest X-rays and predicts eye diseases from retinal fundus images.

User-friendly interface: Easy upload of medical images for risk predictions.

Early warnings: Prompting further medical evaluation for timely intervention.



AIM AND OBJECTIVES

Aim:

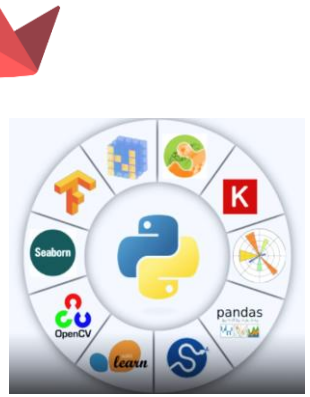
To develop a deep learning tool, MediScan, for proactive disease prediction of tuberculosis and eye diseases (cataracts, glaucoma and diabetic retinopathy), and provide preventive measures using chest X-rays and retinal images.

Objectives:

- Utilize deep learning (CNNs) to create a model for TB and eye disease prediction.
- Achieve high accuracy compared to traditional methods.
- Design a user-friendly web interface for accessibility and self-assessment.
- Contribute to early disease detection, particularly in resource-limited settings.
- Empower individuals to make informed decisions regarding their health.

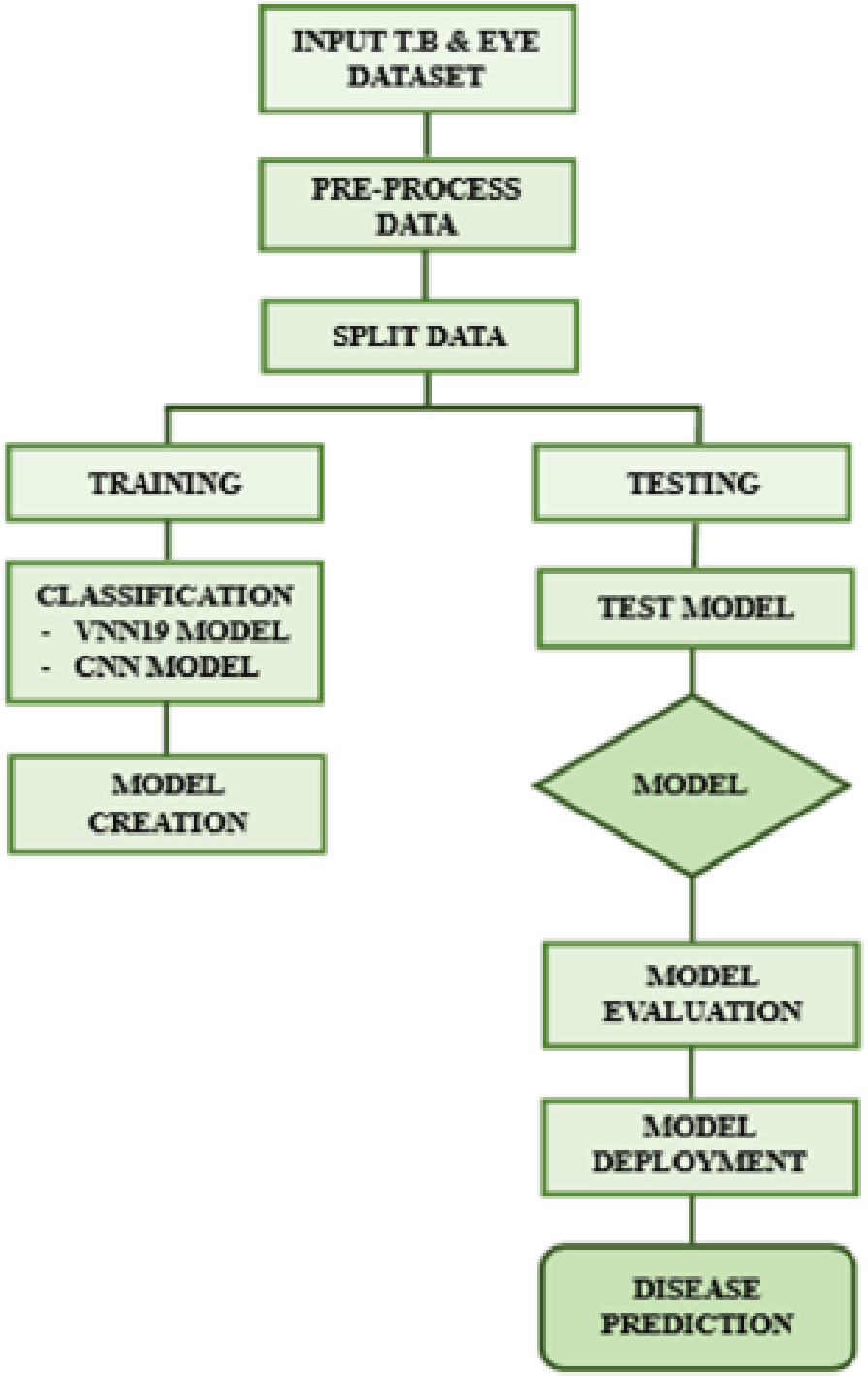
MEDISCAN DEVELOPMENT MATERIALS

- **Data Acquisition:**
  KAGGLE (datasets for model training)
- **Model Building:**
  GOOGLE COLABORATORY (cloud-based development environment)
- **Software Environment:**
  ANACONDA (Python package manager)
- **Programming Language:** PYTHON
- **Development Tools:**
  SPYDER IDE - for efficient coding
  STREAMLIT (web framework) - for user-friendly web app design
- **Front-end Enhancements:**
  BOOTSTRAP (CSS framework) - for potential UI customization



METHODOLOGY

1. **DATA ACQUISITION:** Secured medical image datasets from Kaggle.
2. **DATA PRE-PROCESSING:** Cleaned and prepared data for training.
3. **MODEL BUILDING:** Designed a deep learning architecture.
4. **MODEL TRAINING & EVALUATION:** Trained the model & assess its performance.
5. **DEPLOYMENT:** Integrated the model into a user-friendly web app (Streamlit).
6. **USER INTERFACE DESIGN:** Developed a user-friendly interface for image upload and risk predictions.



DEEP LEARNING CRUCIAL FOR MEDISCAN'S SUCCESS

Optimizing Medical Image Analysis:

- Deep learning algorithms in MediScan analyze medical images with high accuracy and efficiency.
- They can identify subtle abnormalities that might escape human eyes, leading to more reliable diagnoses.

Benefits in healthcare:

- Deep learning accelerates disease prediction from medical images.
- This translates to faster treatment initiation, reduced wait times, and enhanced efficiency.

MEDISCAN BENEFITS: EMPOWERING INDIVIDUALS AND SOCIETY

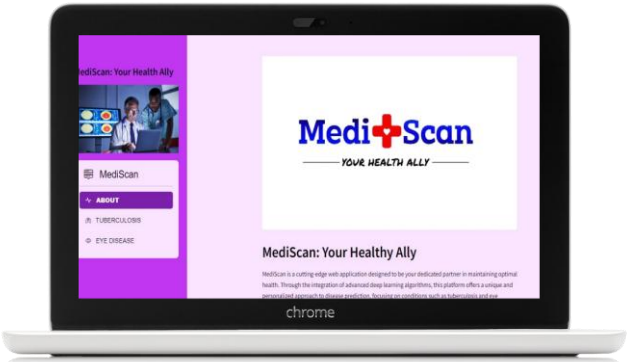
1. Improved Health Outcomes
2. Reduced Financial Stress
3. Lowered Healthcare Costs
4. Increased Economic Productivity
5. Reduced Health Disparities
6. Transformation Potential

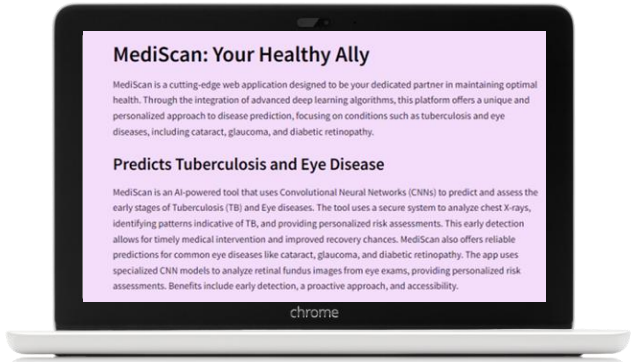
FUTURE ASPECTS

Future Enhancements

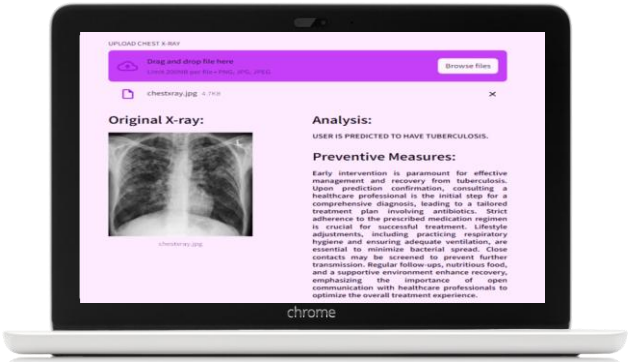
- Integrate MediScan with existing healthcare systems for seamless adoption.
- Expand disease detection capabilities to a wider range.
- Refine model performance, address bias, and enhance interpretability.
- Study real-world impact and collaborate with healthcare providers.

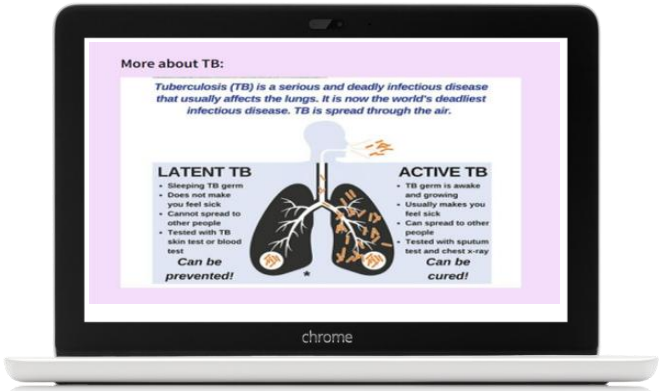
RESULTS


MediScan: Your Health Ally
MediScan
ABOUT
TUBERCULOSIS
EYE DISEASE
Medi Scan
YOUR HEALTH ALLY
MediScan: Your Healthy Ally
chrome


MediScan: Your Healthy Ally
MediScan is a cutting-edge web application designed to be your dedicated partner in maintaining optimal health. Through the integration of advanced deep learning algorithms, this platform offers a unique and personalized approach to disease prediction, focusing on conditions such as tuberculosis and eye diseases, including cataract, glaucoma, and diabetic retinopathy.
Predicts Tuberculosis and Eye Disease
MediScan is an AI-powered tool that uses Convolutional Neural Networks (CNNs) to predict and assess the early stages of Tuberculosis (TB) and Eye diseases. The tool uses a secure system to analyze chest X-rays, identifying patterns indicative of TB, and providing personalized risk assessments. This early detection allows for timely medical intervention and improved recovery chances. MediScan also offers reliable predictions for common eye diseases like cataract, glaucoma, and diabetic retinopathy. The app uses specialized CNN models to analyze retinal fundus images from eye exams, providing personalized risk assessments. Benefits include early detection, a proactive approach, and accessibility.
chrome

MediScan: Your Health Ally
MediScan
ABOUT
TUBERCULOSIS
EYE DISEASE
Tuberculosis Detector
TUBERCULOSIS
chrome


UPLOAD CHEST X-RAY
Drag and drop file here
Browse files
chestxray.jpg
Original X-ray:
Analysis:
USER IS PREDICTED TO HAVE TUBERCULOSIS.
Preventive Measures:
Early intervention is paramount for effective management and recovery from tuberculosis. Upon prediction confirmation, consulting a healthcare professional is the initial step for a comprehensive diagnosis, leading to a tailored treatment plan involving antibiotics. Strict adherence to the prescribed medication regimen is crucial for successful treatment. Lifestyle adjustments, including practicing respiratory hygiene and ensuring adequate ventilation, are essential to minimize bacterial spread. Close contacts may be screened to prevent further transmission. Regular follow-ups, nutritious food, and a supportive environment enhance recovery, emphasizing the importance of open communication with healthcare professionals to optimize the overall treatment experience.
chrome


More about TB:
Tuberculosis (TB) is a serious and deadly infectious disease that usually affects the lungs. It is now the world's deadliest infectious disease. TB is spread through the air.
LATENT TB
Sleeping TB germ
Does not make you feel sick
Cannot spread to other people
Tested with TB skin test or blood test
Can be prevented!
ACTIVE TB
TB germ is awake and growing
Usually makes you feel sick
Can spread to other people
Tested with sputum test and chest x-ray
Can be cured!
chrome

MediScan: Your Health Ally
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TUBERCULOSIS
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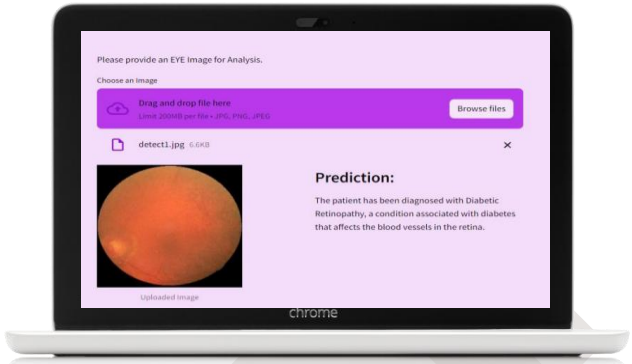
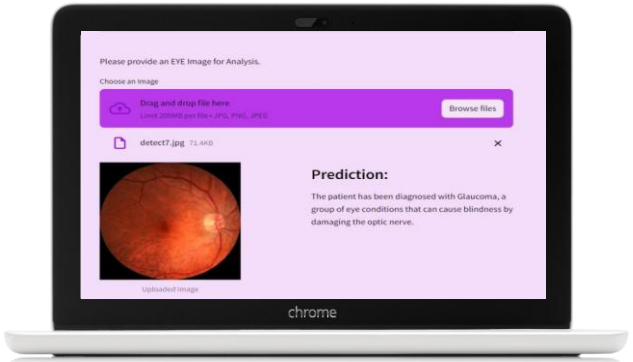
Eye Disease Detection
Eye Disease Predictor
chrome


Please provide an EYE Image for Analysis.
Choose an image
Drag and drop file here
Browse files
detect1.jpg
Prediction:
The patient has been diagnosed with Diabetic Retinopathy, a condition associated with diabetes that affects the blood vessels in the retina.
Uploaded Image
chrome

Cataract Treatment:
CATARACT
chrome


Please provide an EYE Image for Analysis.
Choose an image
Drag and drop file here
Browse files
detect7.jpg
Prediction:
The patient has been diagnosed with Glaucoma, a group of eye conditions that can cause blindness by damaging the optic nerve.
Uploaded Image
chrome

CONCLUSION

- MediScan shows promise in early detection of T.B. and eye diseases.
- Superior accuracy and co-occurrence identification lead to earlier diagnoses and better treatment plans.
- User-friendly interface empowers individuals and potentially improves healthcare efficiency.

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THANK YOU

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